SPREAD: Sampling-based Pareto front Refinement via Efficient Adaptive Diffusion

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pip install moospread
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Challenges in Multi-Objective Optimization (MOO)

Conflict & Trade-offs
Many real-world problems require optimizing multiple conflicting objectives simultaneously.

The Pareto Goal :
The primary goal is to identify the Pareto front, a set of non-dominated solutions where improving one objective deteriorates at least one other.

Scalability & High-Dimensionality
Traditional evolutionary or scalarization methods often struggle with scalability as the dimensionality of the decision and objective spaces grows.



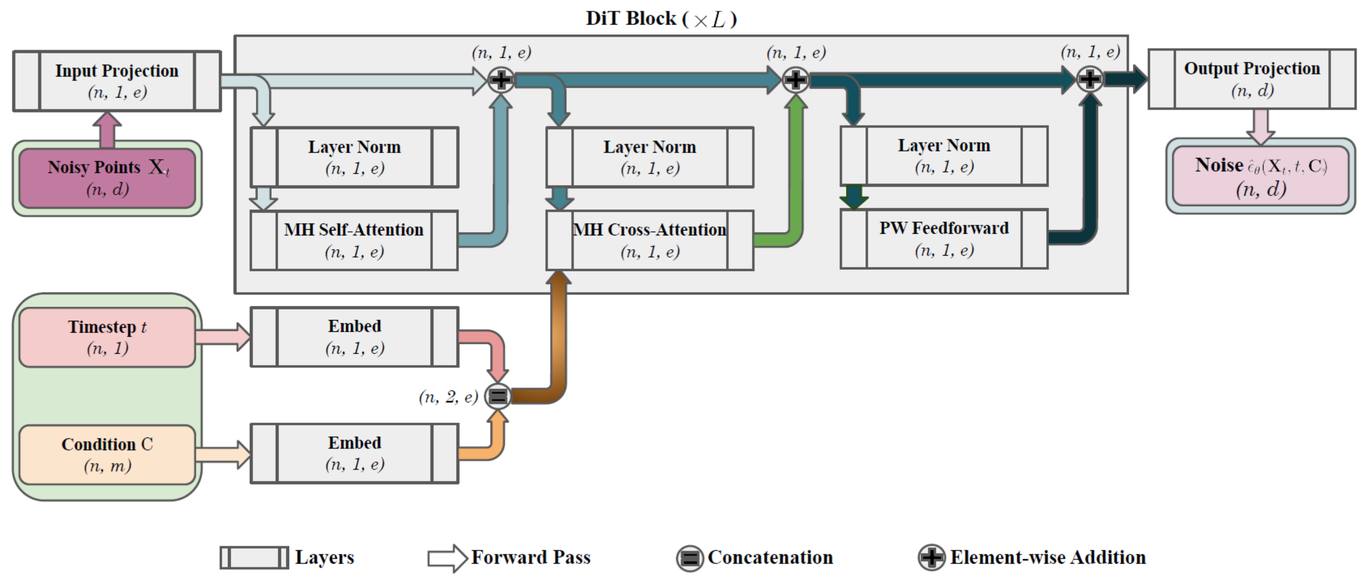
Resource Constraints
There is a critical need for universal frameworks that can adapt to different settings, including Online (direct access), Offline (static pre-collected data), and Bayesian (expensive function evaluations) MOO.



SPREAD Framework & Architecture

SPREAD is a generative framework leveraging Denoising Diffusion Probabilistic Models (DDPMs) to iteratively refine candidate solutions.

Diffusion Transformer (DiT-MOO)
The core is a Diffusion Transformer designed for stable, scalable sampling in optimization contexts.
Input: Batch of n noisy decision variables X_t, timestep t, and condition C.
Process: The model predicts the noise at each reverse step, guiding points toward the latent optimal manifold.



Why Diffusion for MOO?

Iterative Refinement
Diffusion models improve samples step-by-step through a learned reverse process, which naturally aligns with the goal of moving candidate solutions toward the Pareto front.

Conditional Generation
They can learn a conditional process over the decision space, allowing the model to steer sampling toward promising trade-off regions based on specific objective targets.

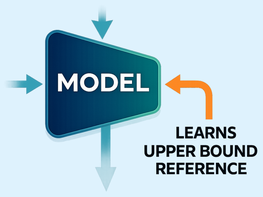
Optimization-Aware Sampling
The reverse diffusion process can be explicitly guided to simultaneously improve objective values and preserve solution diversity.

The Mechanism: Why x_0 Dominates x_T?

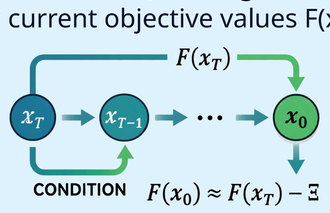
The primary theoretical innovation is Shifted Conditioning, which enables improvement even without active guidance.

Objective Improvement

Training (Shifted Condition): The model is trained on pairs (x, c), where the condition c = F(x) + E (E > 0).



Sampling (Direct Condition): During inference, we sample using the *true* current objective values $F(x_T)$ as the condition.



Result: x_0 dominates x_T with high probability: $F(x_0) < F(x_T)$.

The Logic: The training approach enforces that the distribution associated with a condition c is composed of points whose objective values are approximately c - E (E > 0). When you ask the model to sample using the current objective $F(x_T)$ as the condition, it pulls samples from its learned distribution where the objective values are approximately $F(x_T)$ - E.

Adaptive Guidance: Convergence & Diversity

To move beyond simple improvement to true Pareto optimality, SPREAD applies an Adaptive Multiple Gradient Descent-inspired update.

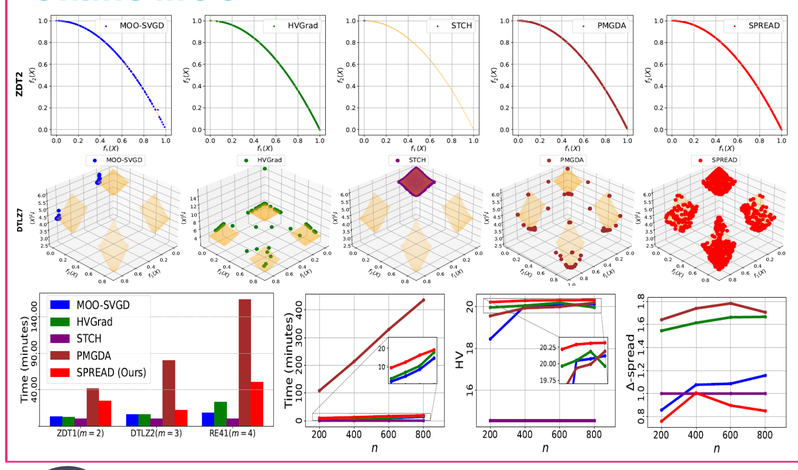
MGD-Guidance
Aligns guidance directions with common descent directions to ensure rapid convergence to the Pareto front.

RBF Repulsion
A Gaussian RBF-based repulsion term discourages clustering and prevents mode collapse, promoting a diverse spread across the front.

Final Solution Set
Maintains the top non-dominated candidates using crowding distance across the entire diffusion trajectory.

Experimental Results

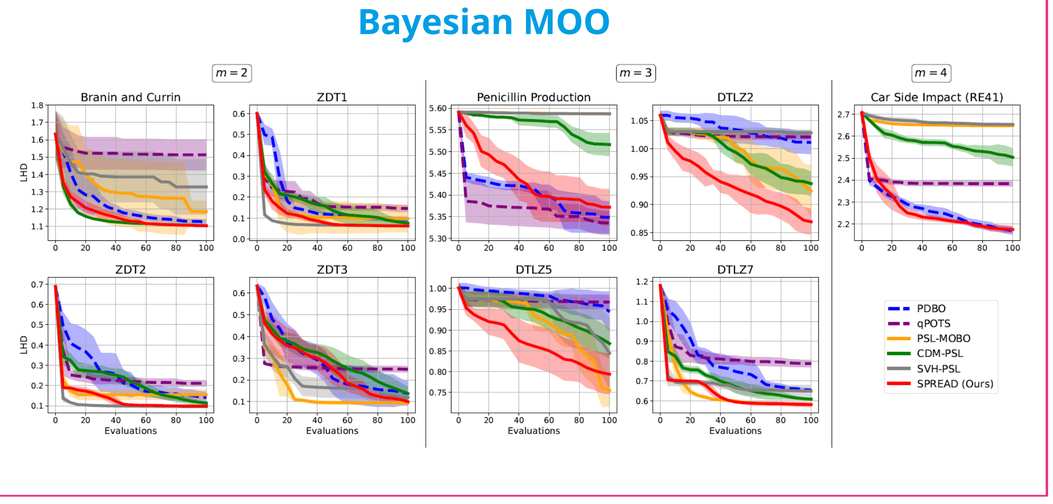
Online MOO



Offline MOO

Method	Synthetic	RE
$\mathcal{D}$(best)	9.08	11.83
MM	5.92	3.92
MM-COM	8.00	7.42
MM-IOM	5.67	4.33
MM-ICT	6.50	3.83
MM-RoMA	6.08	7.75
MM-TriMentoring	8.17	4.58
MH	7.58	5.67
MH-PcGrad	5.75	6.92
MH-GradNorm	9.58	11.50
ParetoFlow	7.50	6.83
PGD-MOO	4.58	8.75
SPREAD	**3.50**	**1.83**

Bayesian MOO



Acknowledgement
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